HDOC-6(EXPERIMENT)

- Q.1-A square photo has side `s` centimeters. A border of width `w` centimeters is added around it. Write a C program to find the total area after adding the border.

ALGORITHM/ PSEUDOCODE:

1. Start
2. Declare the variables s, w, newside, area
3. read the values of s and w
4. Calculate the newside = s + (2 x w)
5. Calculate area = newside x newside
6. Display the total area
7. STOP

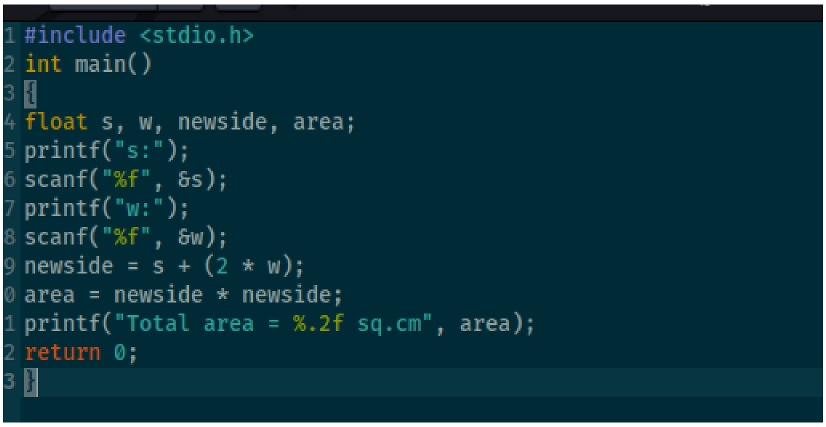
Test case	Input(s, w)	Expected output	Actual output	Result
1.	20,5	900 sq.cm	900 sq.cm	Pass
2.	15,5	625 sq.cm	625 sq.cm	Pass
3.	10,2	196 sq.cm	196 sq.cm	Pass

The code that we are going to use

```
#include <stdio.h>

Int main()

{

float s, w, newside, area;

printf(“s:”);

scanf(“%f”, &s);

printf(“w:”);

scanf(“%f”, &w);

newside = s + (s * w);

area = newside * newside;

printf(“Total area = %.2f sq.cm”, area);

return 0;

}
```

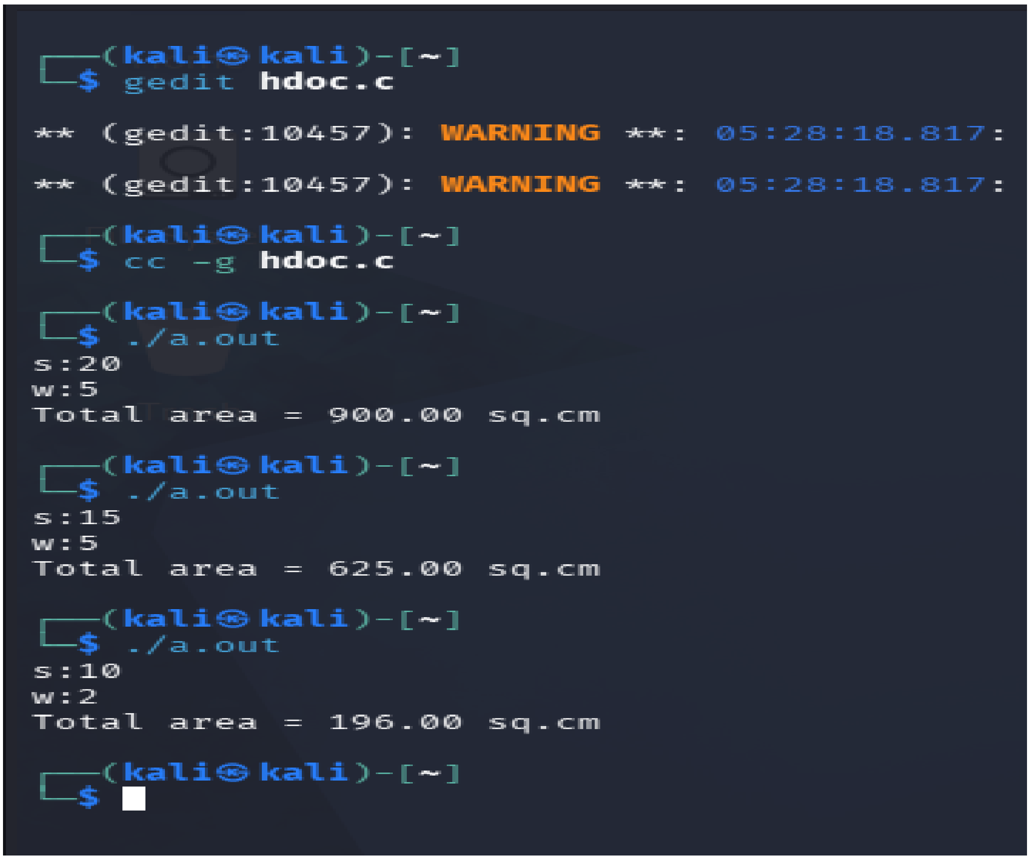
Execution of test cases

The source code



```
#include <stdio.h>
int main()
{
float s, w, newside, area;
printf("s:");
scanf("%f", &s);
printf("w:");
scanf("%f", &w);
newside = s + (2 * w);
area = newside * newside;
printf("Total area = %.2f sq.cm", area);
return 0;
}
```

Answer of the three cases



Now that we have made the algorithm the test case table and executed the code we can move onto the second question which is

- Q.2 Write a C program to find the fencing cost, area, and perimeter of a trapezium-shaped field when all sides, height, and fencing rate per meter are given.

Algoritm/Pseudocode

1. Start

2. Declare variables a, b, c, d, h, rate, area, perimeter, cose

3. Read the values of a, b, c, d, h and rate

4. Calculate area = ((a + b) x h) / 2

5. calculate perimeter = a + b + c + d

6. calculate cost = perimeter x rate

7. display area, perimeter and fencing cost

8. stop

Here:

a = first parallel side

b = second parallel side

c = side1

d = side 2

h = height

TEST CASE TABLE:

Test case	Input(a, b ,c, d, h, rate	Expected output	Actual output	Result
1	10, 8, 6, 9, 6, 110	Area = 54 sq.cm, perimetr = 33m, fencing cost = 3630	same	Pass

2	12, 7, 8, 10, 7, 150	Area = 66.5 sq.cm, perimeter = 37m, fencing cost = 5550	same	Pass
3	14, 9, 10, 12, 9, 200	Area = 103.5 sq.cm Perimeter = 45 m, fencing cost = 9000	same	pass

The code we are going to use

```
#include <stdio.h>

Int main()

{

float a, b, c, d, h, rate;

float area, perimeter, cost

print("a:");

scanf("%f", &a);

printf("b:");

scanf("%f", &b);

printf("c:");

scanf("%f", &c);

printf("d:");

scanf("%F", &d);

printf("h:");

scanf("%f", &h);

printf("rate:");
```

```
scanf("%f", &rate);

area = ((a + b) * h) / 2;

perimeter = a + b + c + d;

cost = perimeter * rate;

printf("Area = %.2f sq.m", area);

printf("perimeter = %.2f m", perimeter);

printf("fencing cost = Rs. %.2f", cost);

return 0;

}
```

Execution of the three test given in the table

```c
#include <stdio.h>

int main()
{
    float a, b, c, d, h, rate;
    float area, perimeter, cost;

    printf("a: ");
    scanf("%f", &a);

    printf("b: ");
    scanf("%f", &b);

    printf("c: ");
    scanf("%f", &c);

    printf("d: ");
    scanf("%f", &d);

    printf("h: ");
    scanf("%f", &h);

    printf("rate: ");
    scanf("%f", &rate);

    area = ((a + b) * h) / 2;
    perimeter = a + b + c + d;
    cost = perimeter * rate;

    printf("Area = %.2f sq.m", area);
    printf("Perimeter = %.2f m", perimeter);
    printf("Fencing Cost = Rs. %.2f", cost);

    return 0;
}
```

```
┌──(kali㉿kali)-[~]
└─$ gedit hdoc2.c

** (gedit:34646): WARNING **: 06:16:59.290: Could not load Peas rep

** (gedit:34646): WARNING **: 06:16:59.290: Could not load PeasGtk

┌──(kali㉿kali)-[~]
└─$ cc -g hdoc2.c

┌──(kali㉿kali)-[~]
└─$ ./a.out
a: 10
b: 8
c: 6
d: 9
h: 6
rate: 110
Area = 54.00 sq.mPerimeter = 33.00 mFencing Cost = Rs. 3630.00

┌──(kali㉿kali)-[~]
└─$ ./a.out
a: 12
b: 7
c: 8
d: 10
h: 7
rate: 150
Area = 66.50 sq.mPerimeter = 37.00 mFencing Cost = Rs. 5550.00

┌──(kali㉿kali)-[~]
└─$ ./a.out
a: 14
b: 9
c: 10
d: 12
h: 9
rate: 200
Area = 103.50 sq.mPerimeter = 45.00 mFencing Cost = Rs. 9000.00

┌──(kali㉿kali)-[~]
└─$ gedit hdoc2.c

** (gedit:37915): WARNING **: 06:22:47.280: Could not load Peas rep

** (gedit:37915): WARNING **: 06:22:47.280: Could not load PeasGtk

┌──(kali㉿kali)-[~]
└─$
```